Data Semantics - Assignment 3 - Generate with Transformers

Riccardo Pozzi

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Overview

This assignment focuses on autoregressive generative models. You will work directly with the `openai-community/gpt2` model and manually implement a `generate` function. This function will generate tokens one at a time, autoregressively, either until a predefined maximum number of new tokens has been generated, or until a stop token is encountered.

You are provided with a Jupyter notebook[1] that contains the base code required for the exercises. The notebook includes useful hints, partial implementations, and expected outputs to guide your work.

You may choose to run the notebook either locally by installing Jupyter[2], or in the cloud using Google Colab.

Questions

Run the first cells of the provided notebook and answer the following questions about model's output:

1. What does `output.logits` represent?
2. What does `output.logits[0, -1, :]` represent?

Exercises

Complete the following tasks in the notebook:

1. Implement the greedy generation strategy by completing the `greedy_generate` function. Greedy decoding should always choose the most probable next token at each step. Refer to the notebook for expected output.

[1] `https://drive.google.com/file/d/16954hYZTvYSfpRxIwSy_oxbBfTQg0MEN/view?usp=sharing`

[2] `https://jupyter.org/install`

2. Extend the generation function to support random sampling from the token distribution, instead of always picking the most probable token.

3. Implement `top_k` sampling, where the next token is randomly selected from the top-k most probable tokens.